

ITAI 1371 Introduction to Machine Learning

Professor Viswanatha Rao (Vishwa)

Assignment: Lab 06 Exploratory Data Analysis

Group Member Contributions

Monepha Mitchell

My individual contribution to this lab included completing and reviewing both required machine-learning tasks. I trained the Linear Regression model to predict passenger fares using age and passenger class, generated predictions, and evaluated the model with mean squared error and root mean squared error. I also prepared the classification data, trained the Logistic Regression model to predict passenger survival, and evaluated its performance with accuracy. In addition, I added comments to explain the main coding steps and answered the knowledge-check questions in my own words. One important takeaway from this lab was that the evaluation metric must match the type of problem: regression metrics measure numerical prediction errors, while classification metrics measure how correctly the model assigns class labels.

Ved Dalwadi

For Module 6, I ran both models and learned about the difference between linear regression and logistic regression. Which for linear regression, I was given code that predicted the fare based on age and Pclass and evaluated using mean squared error. While the logistic regression model was used to predict passenger survival using age Pclass and sex. I helped my teammates better understand why some features were used but not all or just 1 as 1 is too little and all would be too many, and it's important to pick the ones that are only important to the specific thing you're looking for.

Nigel Bontif,

For Module 06, I contributed by working through the Regression and Classification lab notebook and helping complete the model-building sections. I worked on the Linear Regression part for predicting passenger fare and the Logistic Regression part for predicting survival. I also helped clean up the code structure so the cells would run correctly, reviewed the model outputs, and completed the knowledge check answers. My main focus was making sure I understood and explained the difference between regression as predicting a number and classification as predicting a category.